

Md Wazih Yeaser Tanzim

📞 01726102304 ✉ yeasertanzim079@gmail.com [in linkedin.com/wazihyeaser](https://www.linkedin.com/wazihyeaser) github.com/wazihyeaser

Education

Bangladesh University of Engineering and Technology

Graduated July 2024

*Bachelor of Science in Electrical and Electronic Engineering (GPA: 3.** / 4.00)*

Dhaka, Bangladesh

- **Relevant Coursework:** Wireless Communications, Radar and Satellite Communication, Digital Signal Processing I, Digital Signal Processing II, Random Signals and Processes, Continuous Signals and Linear Systems, Communication Systems I, Communication Systems II, Control System I, Microprocessors and Embedded Systems, Computer Networks.

Undergraduate Thesis

Deep Learning Aided Hybrid Beamforming Matrix Estimation for MU-MIMO Terahertz Communication System

Advisor: Prof. Md Forkan Uddin, Bangladesh University of Engineering and Technology.

- Developed a hybrid beamforming matrix estimation technique using deep learning for a multi-user MIMO Terahertz communication system.
- Simulated system performance and demonstrated a significant improvement in spectral efficiency and reduction in computational time complexity compared to conventional approaches.
- Implemented the system using PyTorch and tested it on realistic datasets for wireless communication.

Job Experience

AI Engineer

May 2025 – Present

Shottify

Dhaka, Bangladesh

- Built an LLM-aided news-verification platform with automated large-scale source collection, content normalization, vectorization, AI-driven sentiment analysis, and fake-news / misinformation detection to assess veracity, political bias, and credibility at scale.

Research Assistant

June 2025 – June 2026

Committee for Advanced Studies and Research (CASR)

Dhaka, Bangladesh

Bangladesh University of Engineering and Technology (BUET)

- Proposed deep-unrolled digital beamforming, DL-aided hybrid beamforming, and federated-learning frameworks for cell-free massive MIMO systems under max-min fairness objectives, leveraging DeepMIMO ray-traced data and a physics-inspired, channel-aware aggregation scheme.

Research Assistant

November 2024 – May 2025

Supervised by Dr. Lutfa Akter

Dhaka, Bangladesh

Bangladesh University of Engineering and Technology (BUET)

- Implemented beamforming algorithms (alt-min, OMP, fully-digital) and convex-optimization solvers in MATLAB and Python, reproducing several benchmark cell-free massive MIMO papers and foundational wireless-communication concepts.

Research Interest

Wireless Communication Systems: 6G and Beyond, Cell-Free Massive MIMO, Hybrid and Digital Beamforming, MIMO and Channel Estimation, Reconfigurable Intelligent Surfaces (RIS), Fluid, Movable, and Pinching Antenna Systems, Holographic MIMO, Near-Field Communications, Terahertz (sub-THz) Communications, Integrated Sensing and Communication (ISAC), Deep Learning for the Physical Layer, and Semantic Communications.

Machine Learning & Signal Processing: Leveraging machine learning and deep learning—including model-based and physics-informed (deep-unrolled) methods—to advance signal processing across wireless communications, image processing, and speech recognition.

Current Research Activities

Deep-Unrolled Max-Min Precoding for Cell-Free Massive MIMO

Manuscript in preparation (IEEE TWC)

- Developing an unsupervised deep-unrolling network that maps estimated CSI to max-min-fair precoders for cell-free massive MIMO, trained label-free on a worst-user spectral-efficiency objective.
- Reformulating the iterative max-min / LP-MMSE solver as a differentiable unrolled architecture (regularized-inverse direction → projected-gradient SE ascent → max-min power allocation).

Deep Learning for Hybrid Beamforming in Multi-User Cell-Free MIMO

In progress

- Designing a deep-learning precoder that predicts hybrid (analog phase-shifter + digital) beamformers directly from estimated CSI under the non-convex unit-modulus constraint.
- Targeting worst-user (max-min) spectral efficiency in multi-user cell-free MIMO.

Federated Learning for Distributed Cell-Free MIMO across mmWave Sites

In progress

- Developing a federated-learning framework for privacy-preserving precoder training across multiple city-scale 28 GHz sites (DeepMIMO-based multi-site pipeline).
- Investigating channel-aware model aggregation that accounts for inter-site channel diversity across distributed access points.

Deep Learning for Bangla Speech Processing: Speaker Recognition and Dialectal ASR

In progress

- Developing a deep-learning framework for automated Bangla speaker recognition and an end-to-end speech-to-text system targeting regional dialects, addressing accent variation underrepresented in standard Bangla ASR.
- Focusing on robustness to low-resource dialectal data and real-world acoustic conditions to improve speaker identification and transcription accuracy across Bangla-speaking regions.

Undergraduate Projects

ML-based power system fault detection, classification and location estimation. | MATLAB, Simulink, MS Excel

- Built an ML pipeline on a SIMULINK-simulated dataset to detect and classify the three primary power-system fault types and estimate fault locations, enhancing grid reliability and maintenance planning.

Automated Attendance system based on speech and voice recognition using KNN. | MATLAB, Machine Learning, MFCC

- Developed a KNN model with MFCC features that jointly recognizes speech content and speaker identity to automate attendance, achieving reliable multi-task speech and speaker recognition.

Gesture control wheelchair using Arduino | Arduino, Sensors: MPU6050, HC-SR04, Motor Driver

- Built an Arduino-based wheelchair that translates MPU6050 accelerometer hand-gesture input into real-time motor control (forward, backward, turn), improving mobility for users with limited movement.

FPGA-Based Automated Fan Control with Human Presence and Temperature Sensing | Verilog, Arduino, Python

- Designed an FPGA (Verilog) controller that toggles a BLDC/DC fan and modulates its speed from human-presence and temperature sensing, optimizing comfort and energy efficiency.

Technical Strengths

Programming Languages: MATLAB, Python, C++, C, Java, HTML, CSS

Simulation Software: Proteus, Quartus, PSpice, Simulink, Cyme PSAF, Cisco Packet Tracer

ML Related Frameworks: Pytorch, Numpy, Panda, Matplotlib, Seaborn, OpenCV, Tensorflow

Concepts: Compiler, Artificial Intelligence, Machine Learning, Neural Networks, Hybrid Beamforming, MIMO, Terahertz Communications

Utilities: LATEX, Git, Anaconda, Jupyter Notebook, AutoCAD, Kaggle, Google Colab, Microsoft Office, Arduino, Keil uvision, Fieldprogrammable gate array(FPGA)

English Language Proficiency

IELTS

Listening: 8.0

Reading: 8.0

Writing: 6.5

Overall Band: 7.5

Speaking: 6.0